Exclusion of deep vein thrombosis by measuring spot skin temperatures using a hand-held thermo-comparator

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Abstract

Objectives: Thermograms have shown that deep vein thrombosis (DVT) causes an increase in temperature in the leg. We have studied for the first time, the viability of measuring spot skin temperatures using a hand-held instrument called a thermo-comparator as an inexpensive, simple and reliable method of excluding the diagnosis of DVT.
Methods: A total of 33 patients with suspected DVT had spot skin temperatures taken on specified areas of each leg using the instrument. The results were compared with duplex scans of the patient and with venography in some patients.
Results: There was a mean temperature gradient of 3.38°C down each normal leg. All 10 patients with DVT confirmed by duplex scan or venogram also had raised temperatures compared with surrounding areas, without the normal temperature gradient down the leg (true positive, 100%). This instrument did not have any false negatives and achieved 100% sensitivity and 100% negative predictive value.
Conclusions: The thermo-comparator excluded a diagnosis of DVT in 39% of patients and has the potential for use in primary care as a screening tool to reduce the number of patients sent to hospital unnecessarily.

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Keywords: Deep vein thrombosis; thermal imaging; thermo-comparator; screening tool; primary care

Introduction

Deep vein thrombosis (DVT) is a common problem in the community, with an incidence of 1 in 2000 within the general population and 1 in 500 within those aged over 80 years[1]. However, the number of people presenting with a clinical suspicion of DVT is considerably higher. In our locality, serving a population of 360,000, the number of requests for assessment of clinical DVT is constantly rising. In 2001, there were 1600 requests from general practitioners (GPs) for investigation of patients with suspected DVT. But only one-third of these patients actually had DVT, therefore the majority of these patients underwent unnecessary and time-consuming investigations. Needless to say, this has major resource implications. However, if the diagnosis is missed, propagation of thrombus can lead to more extensive DVT and long-term morbidity associated with chronic venous hypertension (oedema, pain, dermatitis and ulceration) and mortality from pulmonary embolism[2]. Hence the use of accurate tests is imperative.

Modalities of investigation of DVT have changed over the last 40 years. In earlier days the diagnosis was largely a clinical one, although it is now recognized that clinical signs are non-specific, extremely

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unreliable[3] or frequently absent[4]. Furthermore, since only one-third of patients sent to hospital with a suspected DVT are actually found to have the condition, a substantive number of patients are subjected to unnecessary investigations. Hence there is a need for a simple screening method to exclude DVT in clinically suspected cases in primary care[5]. Although the recent explosion of technological advances has seen numerous investigative techniques being evaluated for the diagnosis of DVT, none of them are suitable for use as a screening tool in primary care. An ideal investigative tool for screening (Table 1) should have several characteristics that would make its use desirable[6]; however, no such device is available at present.

Thermal imaging

Since its first report in 1972, thermography has been established as an effective method of excluding DVT[7]. Thermal imaging for DVT is undertaken by infrared imaging and by liquid crystal contact thermography. It is a highly sensitive, quick, simple, noninvasive and risk-free initial investigation for suspected DVT[8]. In the normal subject, infrared imaging will demonstrate no significant asymmetry between the two lower limbs on anterior and posterior views[9], and a normal temperature gradient of about 3°C between the upper thighs and lower calves[10,11]. In patients with DVT, there is an increased skin temperature and loss of the normal temperature gradient down the limb[12,13]. Earlier studies using thermography have reported 100% sensitivity for detecting DVT[14,15]. A negative thermogram thus effectively excludes DVT and spares the patient from further unnecessary investigation or treatment.

However, the machinery is bulky and expensive, and hence too cumbersome for routine use or for use in primary care. Furthermore, thermography is nonspecific for DVT because there are a number of other conditions, such as ruptured Baker's (popliteal) cyst, cellulitis, erysipelas, thrombophlebitis, muscle tears, ligament sprains, arthritis and varicose veins, that also cause an increased temperature in the limb[14]. Hence people with a positive thermogram still need a duplex scan or venography for a conclusive diagnosis of DVT.

Table 1 Characteristics required for an ideal screening tool

	Desirable factors		
Essential factors	Patient factors	Operator factors	Other factors
• Accuracy • Reproducibility • Low invasiveness/ toxicity • Low risk of complications	• Accessibility	• Rapid and simple to perform • Ease of interpretation	• Low cost • Portability

Aim

The aim of this study was to incorporate the thermogram technology with our ideal requirements for a screening tool to produce an inexpensive, simple, hand-held device that could be used to measure temperatures at fixed sites on a leg and detect 'hot spots' that could be DVT, or more importantly to exclude DVT if there are no hot spots.

Materials and methods

A thermo-comparator (Figures 1 and 2) is a hand-held instrument that incorporates the thermogram technology and uses a noninvasive infrared sensing mechanism to detect hot spots in patients with lower limb DVT. This instrument was designed and patented by Caretek Limited®, Cowes, UK. It is inexpensive, simple, easy to use and gives temperature readings instantly.

We carried out a prospective study over an 8-week period on 33 patients (21 women, 12 men; mean age: 61 years; range: 21–89) who presented to our vascular diagnostics unit for investigation of suspected DVT. After obtaining informed consent, patients were asked to lie down for 10 minutes with their legs exposed and heels propped up, so as to allow air to circulate behind the calves and thighs. This was to eliminate any artefacts due to clothing, dressings, etc. and to help the legs reach equilibrium with an ambient draught-free room temperature of 15–20°C.

All patients had temperature measurements taken from both legs (66 limbs). The legs were divided into four vertical quadrants: anteromedial, anterolateral, posteromedial and posterolateral. They were then further subdivided into eight transverse segments, four above the knee joint and four below (Figure 3). We measured spot skin temperatures at 32 specified sites in each leg (one in each segment) using the thermo-comparator. Each spot skin temperature measurement took 12 seconds and it took approximately seven minutes to study each leg in detail. These readings were immediately transferred onto a specially designed chart.

Once the measurements were taken, all 33 patients had a duplex scan. The technician performing the scan was blinded to the thermo-comparator



Figure 1 Lateral view of thermo-comparator



Figure 2 Anterior view of thermo-comparator with temperature reading



Figure 3 Anterior (A) and posterior (B) aspects of the legs with spots

readings. In addition to the duplex scan, venography was performed on four patients in whom the duplex scans were inconclusive. The thermo-comparator measurements were compared with the duplex scan and venogram results.

Results

The spot skin temperature recordings from all 33 patients were analysed. There was a mean temperature gradient of 3.38°C (95% confidence interval [CI], 3.01–3.75°C) in legs without DVT, with the temperature gradually dropping from the upper thigh to the lower calf. This is a 'positive temperature gradient' – a feature of normal legs. On the contrary, the temperature gradients observed in patients with DVTs were very different.

Nine patients had DVT confirmed by duplex scan and one patient had it confirmed after venography (duplex scan was inconclusive). These 10 patients (30%) also had a demonstrable increase in temperature in specified spots in the corresponding leg (true positive). Among them, seven patients had a 'reverse or negative temperature gradient' (lower leg higher temperature than the thigh) with a mean of –0.7°C (range: –0.2 °C to –3.2°C), in at least one of the four quadrants in the leg. Two patients had a drop of less than 1°C (compared with the mean drop of 3.38°C in a normal leg) from the upper thigh to the lower calf level in at least one of the quadrants of the leg, thus signifying a raised temperature in the lower leg. One patient had a step rise of 1.8°C between the mid-calf and lower-calf levels (adjacent segments) and this was the site of the thrombus confirmed by duplex scan. All the above 10 patients had calf DVT and the rise in temperature was also over the same anatomical region. None of these patients had any other relevant associated pathologies, such as varicose veins, thrombophlebitis, cellulitis, etc. that might have caused an increase in temperature.

Thirteen patients (39%) who had a negative duplex scan (no DVT) exhibited a normal 'positive temperature gradient' down the leg (true negatives) and 10 patients (30%) with no DVT also had a raised temperature in the lower leg (false positives). None of the patients had DVT without a significant alteration of the normal temperature gradient down the leg or a step rise in temperature between two adjacent segments (no false negatives). The thermo-comparator readings thus gave a 100% sensitivity, 100% negative predictive value, 57% specificity and 50% positive predictive value.

Discussion

DVT is a common but elusive illness that can result in suffering and death if not recognized and treated effectively[16]. However, clinical diagnosis of patients with suspected DVT is notoriously unreliable, as nearly 70% of patients with signs and symptoms of DVT do not actually have DVT on investigation. Therefore, clinical diagnosis alone of suspected DVT might result in unnecessary hospitalization of patients and inappropriate anticoagulation therapy with potentially dangerous consequences. Although various investigative modalities for detecting DVTs have been tried, none of the tests available today could be reliably used in primary care. Since more than two-thirds of the patients sent to hospital with a clinical suspicion of DVT actually turn out to be a false alarm, there is a pressing need for an investigative technique available at the primary care level to exclude the diagnosis of DVT.

Thermograms have previously shown that DVT causes an increase in temperature in the leg[12,13]. This is due to vascular changes mediated by vasoactive amines affecting the capillary circulation, with the capillary veins immediately beneath the dermis dilating sympathetically to carry an increased blood flow around any thrombus obstructing the deep vein. Our instrument detected hot spots in all the patients with a diagnosis of DVT confirmed by duplex scan or venogram. It also detected hot spots in 10 legs without DVT. Among them, one patient had superficial thrombophlebitis, two patients had thrombus in the superficial veins, and in seven patients the cause was unknown. However, these false positives are not a cause for concern because the purpose of this instrument is not to diagnose DVTs, but to exclude the condition. The instrument did not detect any hot spots in the 13 patients who had a negative duplex scan. This was 39% of the total referrals and hence if a GP had used this instrument in primary care then these patients would not have been send to hospital for expensive and time-consuming investigations. Most significantly, none of the patients had DVT without a pronounced alteration in the normal temperature gradient down the leg. Therefore the thermo-comparator did not miss any patient with DVT, thus proving its accuracy and safety.

Our study not only confirms results from previous studies demonstrating a temperature gradient of at least 3°C down a normal leg[10,11] (Figure 4), but also indicates that DVT causes a pronounced reduction and often a complete reversal in normal temperature differences down the affected sectors of the leg (Figure 5). Therefore, it becomes practical

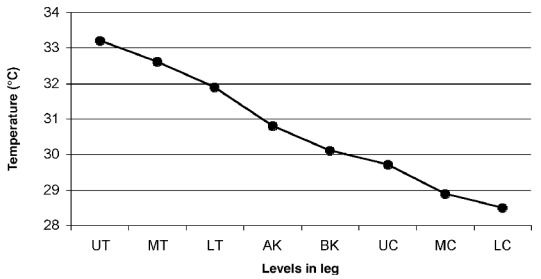


Figure 4 Temperature profile in a normal leg. The temperature drops gradually from the upper thigh down the leg. UT, upper thigh; MT, mid thigh; LT, lower thigh; AK, above knee; BK, below knee; UC, upper calf; MC, mid calf; LC, lower calf

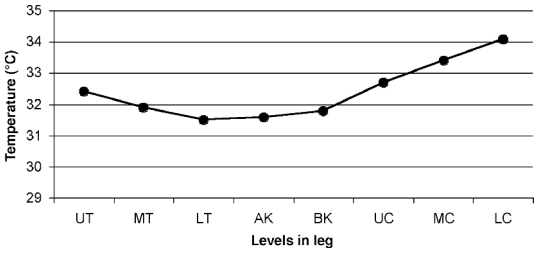


Figure 5 Temperature profile in a leg with calf deep vein thrombosis. The temperature begins to drop from upper thigh to lower thigh, then gradually rises, followed by a significant rise in the region of the calf. This patient had a DVT confirmed by duplex scan. UT, upper thigh; MT, mid thigh; LT, lower thigh; AK, above knee; BK, below knee; UC, upper calf; MC, mid calf; LC, lower calf

to use this instrument to scan a single limb for DVT symptoms without reference to the patient's other limb at all (in amputees or patients with bilateral venous problems).

One potential limitation of this instrument would be the accuracy of assessing DVT in patients with peripheral vascular disease (PVD). This problem could be potentially overcome by detecting a step rise in temperature in adjacent segments of the affected limb that could identify the site of thrombus in addition to a completely altered temperature gradient. However, since none of our series of 33 patients had PVD, further trials including this group of patients need to be done to identify an algorithm and a reliable parameter to be added to the software of the instrument.

Conclusion

In conclusion, the thermo-comparator correctly detected a hot spot on all patients with DVT and also correctly excluded the diagnosis in 39% of patients. This instrument thus has the potential to be used as a screening tool to exclude DVT at a primary care level and thus reduce the number of patients being sent to hospital, as well as for rapid assessment in an accident and emergency department. However, further studies involving a bigger number and different groups of patients need to be done before it can safely be approved for use in primary care. The instrument is currently being refined and developed before further assessment.

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